

ROOSSO P

Backend Engineer

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PROFILE

Python/Django backend engineer specialising in real-time voice AI infrastructure (LiveKit, SIP, Asterisk) and applied generative AI (LoRA/QLoRA fine-tuning for audio and video models). Builds production backend systems end-to-end — from API design through live voice-agent deployment and model evaluation — with direct ownership of debugging and performance issues in real-time telephony pipelines.

TECHNICAL SKILLS

Backend & APIs:	Django, Django REST Framework, REST API design, JWT Auth, Postman
Real-Time & Voice:	LiveKit, SIP Trunking, Asterisk, WebRTC
Frontend:	React.js (Hooks, React Router, Axios), JavaScript, HTML, CSS
Applied AI/ML:	Gemini AI integration, PyTorch, Hugging Face, PEFT, LoRA/QLoRA fine-tuning, ComfyUI
Data:	MySQL, PostgreSQL, MongoDB, SQL
Tooling:	Git, GitHub, Docker, CI-friendly workflows

EXPERIENCE

Python Developer | M7 Corporation Jun 2025 – Present

Voixo AI — AI Calling Platform

- Owned backend API development for an AI-powered calling platform, handling inbound and outbound call flows end-to-end.
- Built and deployed real-time voice agents by integrating LiveKit, SIP trunking, Asterisk, and Gemini AI — including diagnosing and resolving a production room-join race condition (SIGTERM within 500 ms of join) and Deepgram STT misconfiguration in live Tamil-language complaint pipelines.
- Designed API key authentication securing third-party integrations across the platform.
- Debugged sequential call routing failures including SIP 480 DND issues and misconfigured team-lead flags causing calls to be silently bypassed.

Technologies: Python, Django, Django REST Framework, LiveKit, SIP, Asterisk, Gemini AI

Augmented Reality Platform

- Built backend APIs powering an AR content platform, enabling users to generate AR experiences from their own images and video.
- Implemented WebRTC-based real-time video streaming for AR playback and automated AR marker generation using MindAR.js.

Technologies: Django, MindAR.js, WebRTC, JavaScript

Generative AI Fine-Tuning — Ace-Step (Music) & Wan 2.1 (Video)

- Fine-tuned the Ace-Step music generation model using LoRA/QLoRA to improve Tamil and Indian regional music generation quality, including pronunciation accuracy and rhythmic fidelity.
- Fine-tuned the Wan 2.1 video generation and lip-sync model via ComfyUI-based LoRA training, improving lip-sync performance and output video quality across evaluation runs.

Technologies: PyTorch, Hugging Face, PEFT, LoRA, QLoRA, ComfyUI

Math AI Trainer | Outlier AI Sep 2024 – Nov 2024

- Created mathematical reasoning prompts for AI model training and evaluated AI responses to improve logical reasoning and problem-solving quality.

PROJECTS

AnalyView — Django Architecture Visualiser (Personal Project)

- Built a Python package that performs AST-based static analysis of Django projects, resolving cross-app imports, detecting model/view/URL/inheritance relationships, and rendering interactive architecture diagrams in a React Flow frontend.
- Implemented an ImportResolver, SettingsExtractor, AppBoundaryDetector, and SymbolIndex pipeline to map project-wide Python file relationships without runtime execution.
- Packaged and published to PyPI; CLI built with Typer, backend served via FastAPI, frontend uses ERP-style column layout with cross-app edge highlighting.

Technologies: *Python, Django, AST, FastAPI, Typer, React.js, React Flow*

SRS Blog — Full-Stack Blogging Platform (Personal Project)

- Built a full-stack MERN blogging application with JWT authentication, rich-text editing (CK Editor), story search/pagination, likes, reading lists, comments, and image uploads.
- React.js frontend with Hooks and React Router; Node.js/Express backend; MongoDB for persistence; Nodemailer for transactional email.

Technologies: *React.js, Node.js, Express, MongoDB, JWT, Multer, Nodemailer*

Freelance — Saint Mary Magdalene Church Management System

- Delivered a full-stack church management system (React.js frontend, Django backend) including an admin dashboard for content management.

Technologies: *React.js, Django, REST APIs, MySQL*

Other Personal Projects

- **Task Management API:** Secure REST API with JWT auth, CRUD, and role-based access filtering. [GitHub](#)
- **Diabetes Prediction (ML):** KNN classifier for diabetes risk detection from health metrics — 76.6% test accuracy. [GitHub](#)

EDUCATION

B.Tech, Artificial Intelligence & Data Science

Sri Muthukumaran Institute of Technology | 2020 – 2024 | CGPA: 8.1 / 10

CERTIFICATIONS

- **Microsoft Azure AI Essentials** — Microsoft
- **Full Stack Development (Node.js & MongoDB)** — Infosys Springboard
- **Docker Foundations** — LinkedIn Learning
- **Cybersecurity Introduction** — Cisco